C3.2 Read data channel signaling

The following sections define the additional response and acknowledge signaling and constraints on the read data channel. See *Read and write response structure* on page A3-54 for information on the baseline read response signaling.

C3.2.1 Read response signaling

Table C3-14 shows the additional read response signals:

Table C3-14 Additional RRESP read response bits

Signal	Source	Description
RRESP[2]	Interconnect	PassDirty
RRESP[3]	Interconnect	IsShared

RRESP[2], PassDirty bit

When this bit is HIGH, it indicates that the cache line is dirty with respect to main memory and the initiating master must ensure that the cache line is written back to main memory, at some time. The initiating master must either perform the write, or pass the responsibility to perform the write to another master.

When this bit is LOW, it indicates that it is not the responsibility of the initiating master to ensure the cache line is written back to main memory.

RRESP[3], IsShared bit 表示其他的Cache设备中也有这个CacheLine，所以自己的这个CacheLine的权限需要为Shared

When this bit is HIGH, it indicates that another copy of the associated data might be held in another cache and the cache line must be held in a Shared state.

When this bit is LOW, it indicates that this is the only cached copy of the associated data and the cache line can be held in a Unique state.

The IsShared and PassDirty responses have the following restrictions:

- the IsShared and PassDirty responses must be constant for all data transfers within a burst
- the IsShared response must be LOW for the transactions that require all other cached copies to be removed. The transactions that require all other cached copies to be removed are:
  - ReadUnique
  - CleanUnique
  - MakeUnique
  - CleanInvalid
  - MakeInvalid.
- the PassDirty response must be LOW for the transactions that do not permit the passing of dirty data. The transactions that do not permit the passing of dirty data are:
  - ReadOnce
  - ReadClean
  - CleanUnique
  - MakeUnique
  - CleanShared
  - CleanInvalid
  - MakeInvalid.

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- The IsShared and PassDirty response must be LOW for the following transactions for which they have no meaning:
  - ReadNoSnoop
  - Barrier transactions
  - DVM transactions.

The value of **RRESP[3:2]** must be the same for all data transfers in a burst.

The following transactions have a single read data channel transfer:

- CleanUnique
- MakeUnique
- CleanShared
- CleanInvalid
- MakeInvalid
- Barrier
- DVM.

These transactions complete in a single read data channel transfer and must have **RLAST** asserted. **RDATA** can have any value and must be ignored.

The EXOKAY response is only permitted for a ReadNoSnoop, ReadClean, ReadShared or CleanUnique transaction. See *Read and write response structure* on page A3-54.

Table C3-15 shows the permitted IsShared and PassDirty responses for each transaction:

Table C3-15 IsShared and PassDirty permitted responses

Transaction	IsShared	PassDirty
ReadNoSnoop	0	0
ReadOnce	0	0
	1	0
ReadClean	0	0
	1	0
ReadNotSharedDirty	0	0
	0	1
	1	0
ReadShared	0	0
	0	1
	1	0
	1	1
ReadUnique	0	0
	0	1
CleanUnique	0	0
MakeUnique	0	0
CleanShared	0	0
	1	0

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Table C3-15 IsShared and PassDirty permitted responses (continued)

Transaction	IsShared	PassDirty
CleanInvalid	0	0
MakeInvalid	0	0
Read Barrier	0	0
DVM Message	0	0
DVM Complete	0	0

Note

Table C3-15 on page C3-168 only shows the permitted responses on **RRESP[3:2]**. For the permitted responses on **CRRESP** see *Snoop response channel signaling* on page C3-176.

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